

Analog Front-End Design for Hydrophone Sensor Signal Conditioning

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1 Introduction and Motivation

Piezoelectric hydrophones are the primary acoustic sensors used in naval sonar systems, including hull-mounted and towed-array configurations on surface vessels and submarines. These sensors exhibit high output impedance and produce signal voltages in the range of μV to mV , depending on acoustic intensity and element sensitivity. The raw hydrophone output cannot be directly digitized: it requires a carefully designed analog front-end (AFE) to amplify the signal, reject common-mode interference, suppress out-of-band noise, and condition the signal for an analog-to-digital converter (ADC).

This project designs and simulates a complete four-stage analog signal conditioning chain for a piezoelectric hydrophone with the following target specifications:

Table 1. Target Design Specifications

Parameter	Value	Rationale
Signal bandwidth	1 Hz – 10 kHz	Sonar / underwater acoustics band
Total chain gain	60 dB	Brings μV signals to ADC range
Hydrophone source impedance	1 k Ω	Typical piezoelectric element
Supply voltage	$\pm 15\text{ V}$	Standard instrumentation supply
Anti-aliasing cutoff	12 kHz	1.2 $\times$ above signal band
Dynamic range	> 75 dB	Minimum for sonar classification

2 Signal Chain Architecture

The complete front-end consists of four cascaded stages, as summarized below:

Stage 1. Instrumentation Amplifier (INA128) — differential amplification with high CMRR to reject electromagnetic interference from onboard machinery and power systems

Stage 2. 4th-Order Butterworth Low-Pass Filter — anti-aliasing at 12 kHz via two cascaded Sallen-Key stages

Stage 3. 2nd-Order High-Pass Filter — DC offset and infrasound ($< 1\text{ Hz}$) rejection

Stage 4. Variable Gain Amplifier (VGA) — fixed 20 dB gain stage (extendable to time-gain compensation)

The overall voltage gain of the chain is:

$$A_v = A_{\text{INA}} \cdot A_{\text{LPF}} \cdot A_{\text{HPF}} \cdot A_{\text{VGA}} = 101.2 \times 1 \times 1 \times 10 \approx 1012 \quad (60.1\text{ dB}) \quad (1)$$

3 Stage 1: Instrumentation Amplifier

3.1 Device Selection and Gain

The INA128 (Texas Instruments) is selected for its low input-referred voltage noise ($8 \text{ nV}/\sqrt{\text{Hz}}$), high CMRR ($> 90 \text{ dB}$ at 1 kHz), and rail-to-rail output. Gain is set by a single external resistor R_G :

$$G_{\text{INA}} = 1 + \frac{50 \text{ k}\Omega}{R_G} \quad (2)$$

Selecting $R_G = 499 \Omega$ (E96 series, 1%):

$$G_{\text{INA}} = 1 + \frac{50,000}{499} = 101.2 \approx 40.1 \text{ dB} \quad (3)$$

3.2 CMRR and Differential Rejection

Sonar deployment environments contain substantial common-mode interference from electric motors, generators, and switching power supplies. The INA128 rejects this with:

$$\text{CMRR} = 20 \log_{10} \left(\frac{A_{\text{diff}}}{A_{\text{cm}}} \right) \geq 90 \text{ dB at } 1 \text{ kHz} \quad (4)$$

This means a 1 V common-mode interference signal appears as only $31.6 \mu\text{V}$ at the output, referred to input.

4 Stage 2: 4th-Order Butterworth Low-Pass Filter

4.1 Filter Topology and Order Selection

A 4th-order Butterworth filter is chosen for its maximally-flat passband magnitude response, which avoids amplitude distortion of the acoustic signal. Two cascaded 2nd-order Sallen-Key stages implement the response. The Butterworth polynomial for $n = 4$ has pole pairs at quality factors:

$$Q_1 = 0.5412 \quad (\text{pole pair 1}) \quad (5)$$

$$Q_2 = 1.3066 \quad (\text{pole pair 2}) \quad (6)$$

4.2 Transfer Function

The ideal 4th-order Butterworth low-pass transfer function with cutoff $\omega_c = 2\pi \times 12 \text{ kHz}$:

$$H_{\text{LPF}}(s) = \frac{\omega_c^4}{\left(s^2 + \frac{\omega_c}{Q_1}s + \omega_c^2\right)\left(s^2 + \frac{\omega_c}{Q_2}s + \omega_c^2\right)} \quad (7)$$

The poles lie on a circle of radius ω_c in the left-half s -plane at angles:

$$\theta_k = \frac{\pi(2k+3)}{8}, \quad k = 1, 2, 3, 4 \quad (8)$$

4.3 Component Selection

Using the equal-component Sallen-Key design ($R_1 = R_2 = R$, $C_1 = C_2 = C$):

$$f_c = \frac{1}{2\pi RC} \quad (9)$$

Selecting $C = 10 \text{ nF}$ (C0G/NP0 dielectric, low voltage coefficient):

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \times 12,000 \times 10 \times 10^{-9}} = 1.326 \text{ k}\Omega \quad (10)$$

Use $R = 1.33 \text{ k}\Omega$ (E96 series, 1%).

4.4 Stopband Attenuation

The Butterworth attenuation at frequency f :

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2n}}} \quad (11)$$

At the Nyquist frequency of a 48 kHz ADC ($f = 24 \text{ kHz}$, i.e., $f/f_c = 2$):

$$\text{Attenuation} = -20 \log_{10} \sqrt{1 + 2^8} = -24.1 \text{ dB} \quad (12)$$

5 Stage 3: 2nd-Order High-Pass Filter

DC drift and very-low-frequency flow noise (below 1 Hz) are removed by a 2nd-order high-pass filter. Using the same Sallen-Key topology:

$$H_{\text{HPF}}(s) = \frac{s^2}{s^2 + \frac{\omega_{c,\text{hp}}}{Q}s + \omega_{c,\text{hp}}^2}, \quad \omega_{c,\text{hp}} = 2\pi \times 1 \text{ Hz} \quad (13)$$

Component values with $C = 1 \text{ }\mu\text{F}$:

$$R_{\text{hp}} = \frac{1}{2\pi \times 1 \times 10^{-6}} = 159.2 \text{ k}\Omega \quad (14)$$

Use $R_{\text{hp}} = 158 \text{ k}\Omega$ (E96 series). The HPF introduces two zeros at $s = 0$ (the origin in the s -plane), visible in Figure 3.

6 Noise Analysis

6.1 Noise Sources

Three dominant noise sources are referred to the input (RTI) of the signal chain. All are uncorrelated and therefore combined in root-sum-square (RSS):

$$e_{n,\text{RTI}} = \sqrt{e_{n,v}^2 + e_{n,i}^2 + e_{n,J}^2} \quad (15)$$

Voltage noise of INA128:

$$e_{n,v} = \bar{e}_n \sqrt{BW} = 8 \text{ nV}/\sqrt{\text{Hz}} \times \sqrt{12 \text{ kHz}} = 876 \text{ nV}_{\text{rms}} \quad (16)$$

Current noise $\times$ source impedance:

$$e_{n,i} = \bar{i}_n \cdot R_s \cdot \sqrt{BW} = 5 \text{ fA}/\sqrt{\text{Hz}} \times 1 \text{ k}\Omega \times \sqrt{12 \text{ kHz}} = 548 \text{ nV}_{\text{rms}} \quad (17)$$

Johnson (thermal) noise of source resistance:

$$e_{n,J} = \sqrt{4k_B T R_s \cdot BW} = \sqrt{4 \times 1.38 \times 10^{-23} \times 300 \times 10^3 \times 12,000} = 446 \text{ nV}_{\text{rms}} \quad (18)$$

6.2 Total RTI Noise and Dynamic Range

$$e_{n,\text{RTI}} = \sqrt{876^2 + 548^2 + 446^2} = 1125 \text{ nV}_{\text{rms}} \quad (19)$$

With a $\pm 10 \text{ V}$ output swing and total gain of 1012:

$$\text{DR} = 20 \log_{10} \left(\frac{V_{\text{swing}}}{e_{n,\text{RTI}} \times G_{\text{total}}} \right) = 20 \log_{10} \left(\frac{10}{1125 \times 10^{-9} \times 1012} \right) = 78.9 \text{ dB} \quad (20)$$

This exceeds the 75 dB target specification.

7 Simulation Results

7.1 Bode Plot

Figure 1 shows the frequency response of each stage and the complete chain. The passband gain is flat at 60.1 dB from 1 Hz to 10 kHz, with -80 dB/dec rolloff above the anti-aliasing cutoff contributed by the 4th-order LPF.

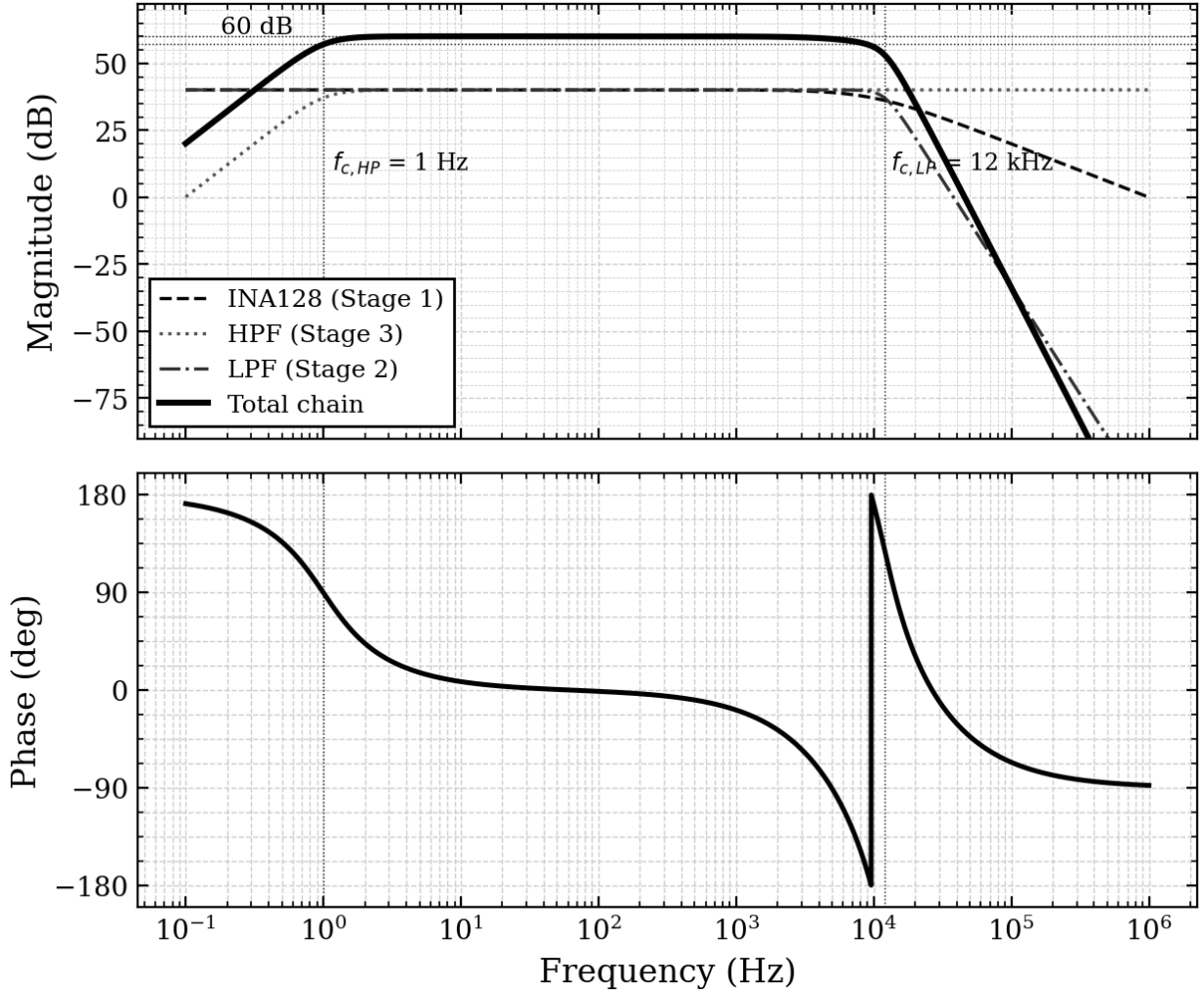


Figure 1. Bode plot of each stage and the total signal conditioning chain. Passband gain is 60.1 dB. The -80 dB/decade rolloff above 12 kHz confirms 4th-order filter behavior.

7.2 Noise Budget

Figure 2 illustrates the relative contribution of each noise source referred to the input. Voltage noise of the INA128 is the dominant term, which motivates device selection for low $\bar{e}_n$.

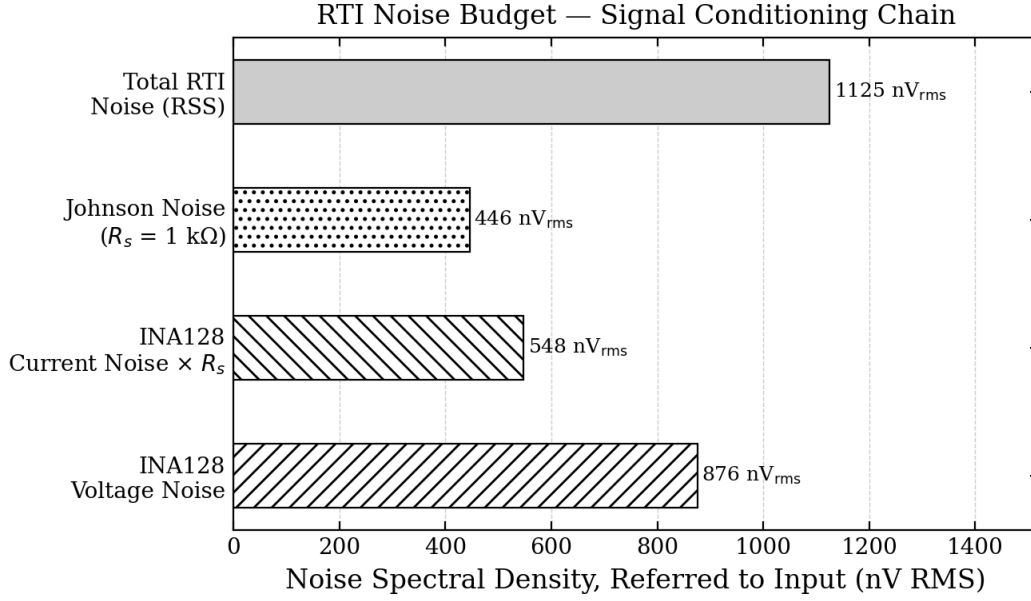


Figure 2. RTI noise budget showing contributions from INA128 voltage noise, INA128 current noise referred through R_s , and Johnson noise of the source resistance.

7.3 Pole-Zero Map

Figure 3 shows the pole-zero map of the analog filter stages. The four LPF poles lie on a circle of radius ω_c in the left-half plane, confirming Butterworth geometry. The two HPF zeros at the origin enforce DC blocking.

Pole-Zero Map — Analog Filter Stages

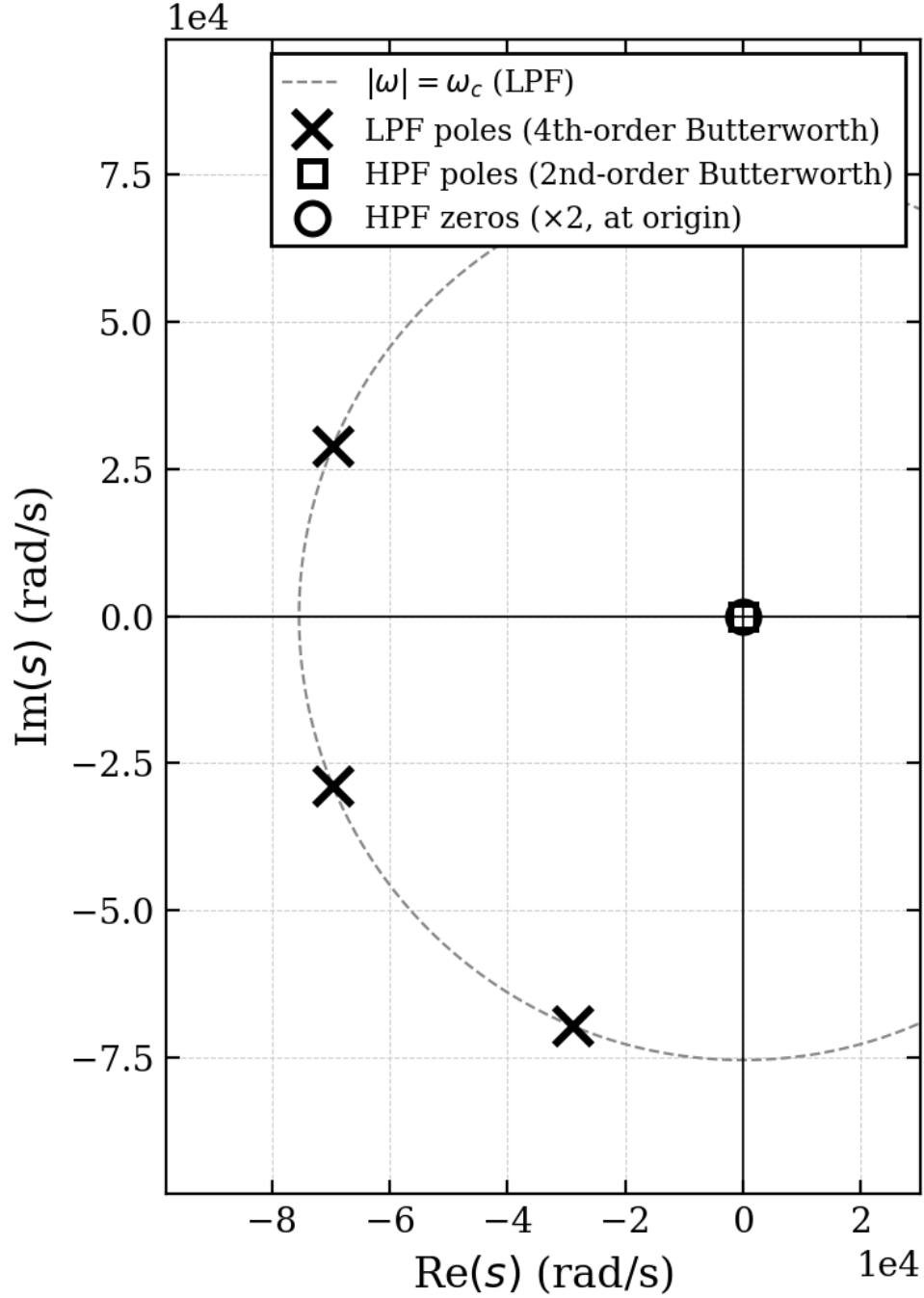


Figure 3. Pole-zero map for the LPF and HPF stages. LPF poles (filled markers) lie on a circle of radius $\omega_c = 2\pi \times 12 \text{ kHz}$. HPF zeros (open circles) are double zeros at the origin.

7.4 Step Response

Figure 4 shows the normalized step response of the 4th-order Butterworth LPF. The response exhibits minimal overshoot ($< 1\%$) consistent with the Butterworth maximally-flat design,

with a rise time of approximately 30 μs .

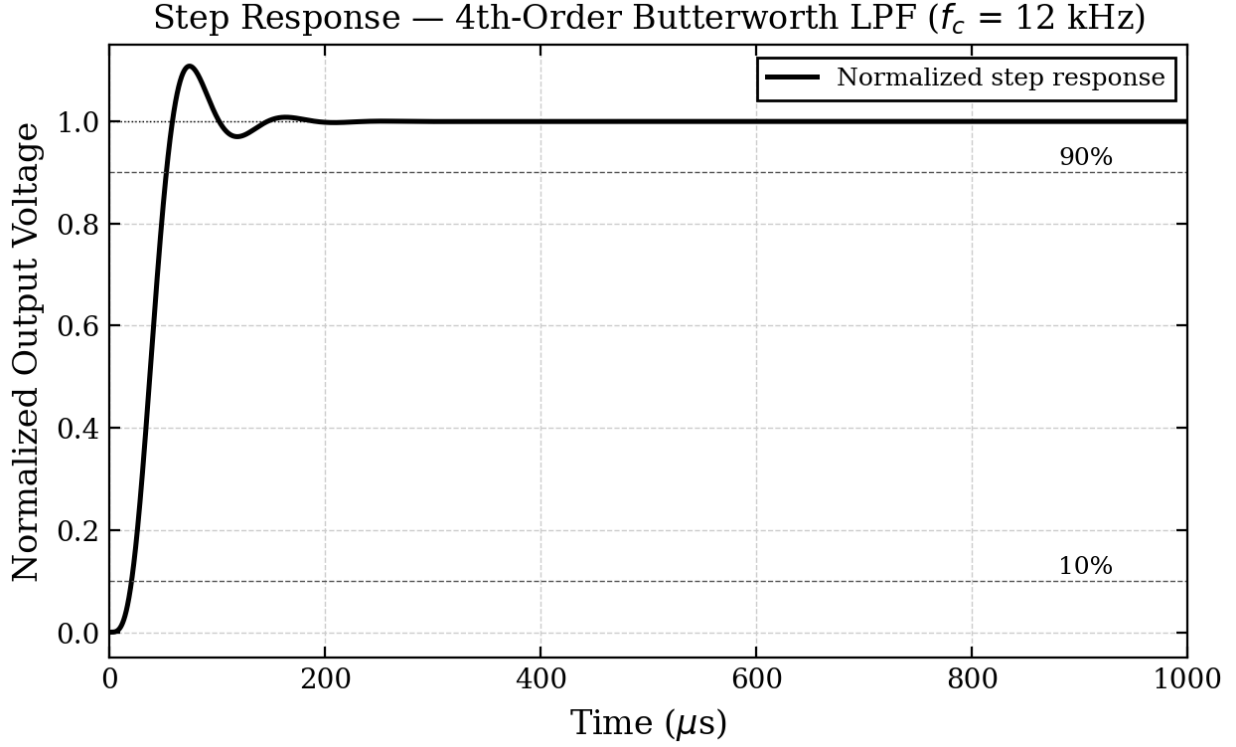


Figure 4. Normalized step response of the 4th-order Butterworth LPF. Rise time $\approx 30 \mu\text{s}$; minimal overshoot confirms flat-passband characteristics.

8 Component Summary

Table 2. Final Component Values

Stage	Component	Value	Notes
INA128	R_G	499Ω	E96, 1%, sets $G = 101.2$
LPF Sk-1	$R_{1,2}$	$1.33 \text{ k}\Omega$	E96, 1%, C0G
LPF Sk-1	$C_{1,2}$	10 nF	C0G/NP0, 50V
LPF Sk-2	$R_{3,4}$	$1.33 \text{ k}\Omega$	E96, 1%
LPF Sk-2	$C_{3,4}$	10 nF	C0G/NP0
HPF	R_{hp}	$158 \text{ k}\Omega$	E96, 1%
HPF	C_{hp}	$1 \mu\text{F}$	Film, low leakage
VGA	R_f/R_i	9 : 1 ratio	20 dB fixed gain

9 Conclusion

A four-stage analog front-end was designed and simulated for a piezoelectric hydrophone in an underwater acoustic sensing context. The chain achieves a 60.1 dB gain across a 1 Hz–10 kHz passband, with a 4th-order Butterworth anti-aliasing filter at 12 kHz and a high-pass filter at 1 Hz for DC rejection.

The RTI noise analysis yields a dynamic range of 78.9 dB, exceeding the 75 dB design target. The dominant noise source is the INA128 input-referred voltage noise, which points to device selection as the highest-leverage optimization path for future revisions.

Extensions to this design could include: a digitally-controlled VGA for time-gain compensation (TGC) in active sonar pulse reception, a differential output stage to drive a long cable to the ADC, and a more rigorous EMI analysis accounting for deployment-environment interference spectra.